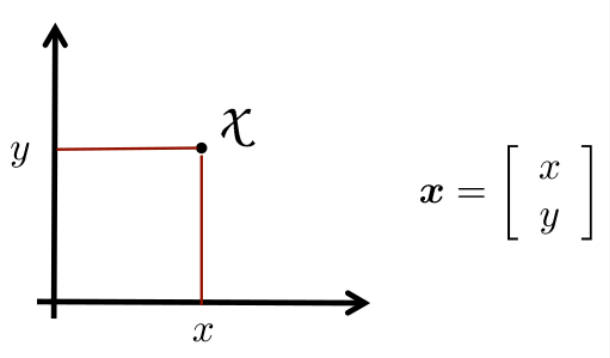
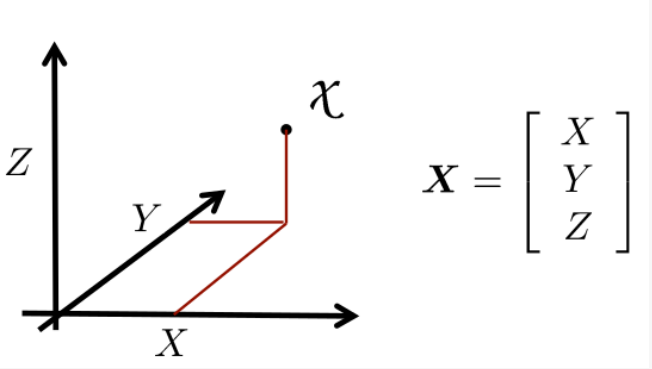
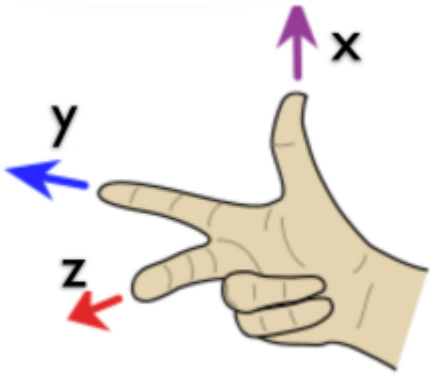
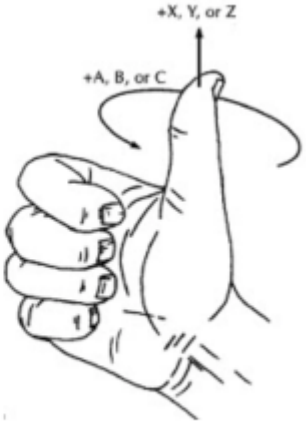
L1-1: Coordinate Tranformations & Representations for Rotations

	
A Point in 2 D	A Point in 3 D

In 3D: Right-Handed CS



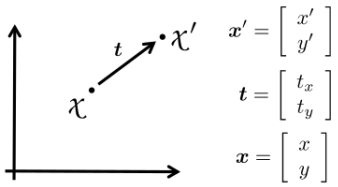
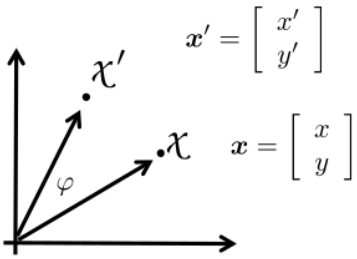
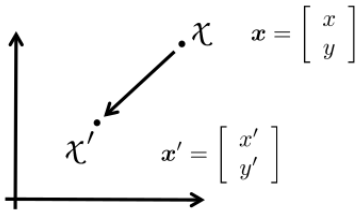
axes



positive rotation

Notation

- **Point** x (or y or p)
  - In Homogeneous coordinates $\mathbf{x}$
  - In Euclidian coordinates $\boldsymbol{x}$
- **2D** vs. **3D** space
  - Lowercase = 2D, Capitalised = 3D
- **Plane** $\mathcal{A}$
  - In homogeneous coordinates $\mathbf{A}$
- **Matrix** R
- **Quaternion** $\mathbf{q}$

Translation	Rotation	Scale Change
		
$x' = \mathcal{T}(x) : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$	$x' = \mathcal{R}(x) : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$	$x' = \mathcal{D}(x) : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \lambda x \\ \lambda y \end{bmatrix}$
Sum	Matrix Multiplication	Matrix Multiplication

Sum vs Matrix Multiplication

- No Single Operation to describe Transformation in Euclidian Space
  - Alternative representation as Quaternions
- Transformations in **Euclidian Space**
  - Combination of multiplication and additions
  - Rotation and Translation are **NOT** commutative
    - $\mathcal{T}(\mathcal{R}(x)) \neq \mathcal{R}(\mathcal{T}(x))$

Similarity Transformation

-	In 2D	In 3D
Scale Change	1 DoF	1 DoF
Rotation	1 DoF	3 DoF
Translation	2 DoF	3 DoF
Sum	4 DoF	7 DoF
-	$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} \lambda x \\ \lambda y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$	$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \lambda R \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_X \\ t_Y \\ t_Z \end{bmatrix}$

Rotations

- Rotation is special transformation

$$\mathcal{R} : \mathbb{R}^n \to \mathbb{R}^N \quad \boldsymbol{x}' = R\boldsymbol{x}$$

- with $|R| = 1$ and $R^{-1} = R^T$
- **Properties**
  - Do not change the scale
  - Have fix point (rotation center)
- **Expression** through:
  - Rotation Matrices
    - Commonly used exchange format for rotations
    - Easy composition of rotations
    - Suboptimal for state estimation problems (9 parameters+constraints)
  - Euler angles
    - Minimal representation
    - useful for visualising rotations
    - Singularities & Discontinuities
    - No direct composition of rotations
    - Suboptimal for general state estimation problems
  - (Normalised) Axis-Angle Representation
    - Singularity for $\theta = 0$
    - Rotations limited to $(-\pi, \pi]$
    - Human readable
    - Normalised: Minimal representation
    - No direct composition of rotation (requires to build rotation matrix)
  - Quaternion
    - 4 Parameter presentation
    - Unique, except of sign
    - no singularities, no discontinuities
    - Easy composition of rotations
    - Allow for angular interpolation (SLERP)
    - Used for state estimation and handling rotations
  - *Matrix exponential of a skew-symmetric matrix*

3D Rotation Parameterisations

Representation	# Parameters	Unique	Readable	Singularities
Rotation Matrix	9	✅	❌	❌

Representation	# Parameters	Unique	Readable	Singularities
Euler Angles	3	❌	✅	✅
Rotation Axis and Angle	4(3)	(❌)	(✅)	(✅)
Quaternion	4(3)	✅	❌	❌

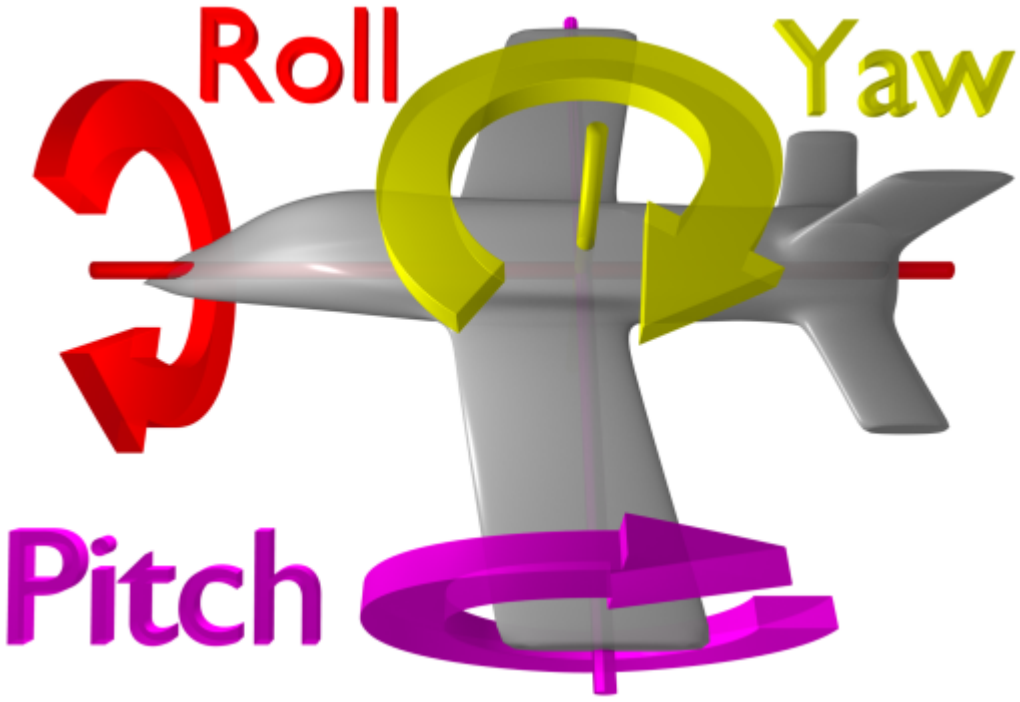
Rotation Matrix

- Rotation of the 2D plane by the angle φ
  - $R(\varphi) = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$
- Rotation in 3D: 3 axes to rotate
  - $R_x(\varphi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \sin\varphi & \cos\varphi \end{bmatrix}$
  - $R_y(\varphi) = \begin{bmatrix} \cos\varphi & 0 & \sin\varphi \\ 0 & 1 & 0 \\ -\sin\varphi & 0 & \cos\varphi \end{bmatrix}$
  - $R_z(\varphi) = \begin{bmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- 9 DoF in general
- 6 independent constraints reduce degrees of freedom (each: -1 DoF)
  - -> Overparameterised representation
- 3 degrees of freedom remain:
  - $\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \left[\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3\right] = \left[\boldsymbol{r}_1^T, \boldsymbol{r}_2^T, \boldsymbol{r}_3^T\right]$
  - Constraints:
    - $|\boldsymbol{c}_1|^2 = 1$
    - $|\boldsymbol{c}_2|^2 = 1$
    - $|\boldsymbol{c}_3|^2 = 1$
    - $\boldsymbol{c}_1^T \boldsymbol{c}_2 = 0$
    - $\boldsymbol{c}_2^T \boldsymbol{c}_3 = 0$
    - $\boldsymbol{c}_3^T \boldsymbol{c}_1 = 0$
- **Properties**
  - Column vectors $\boldsymbol{c}_i$ of R are the **images of the unit vectors** $\boldsymbol{e}_i$

        - $\boldsymbol{c}_i = R\boldsymbol{e}_i = \left[\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3\right] \boldsymbol{e}_i$
        - *Column vector -> previous unit vector*
    - Row vectors $\boldsymbol{r}_i$ of R are the vectors, which have been rotated into the unit vectors $\boldsymbol{e}_i$
    - $\boldsymbol{r}_i^T = \boldsymbol{e}_i^T R = \boldsymbol{e}_i \left[\boldsymbol{r}_1^T, \boldsymbol{r}_2^T, \boldsymbol{r}_3^T\right]$
        - *Row vector -> what is now a unit vector*

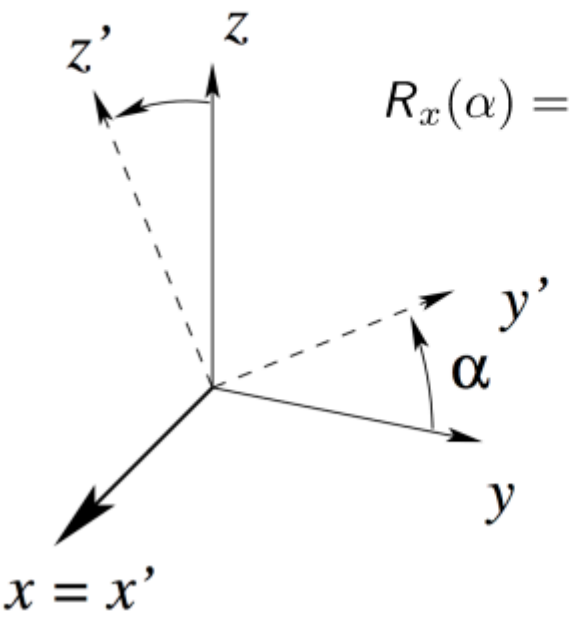
Euler Angles

- 3 rotations around fixed axes (e.g. z-y-x axes)
- Useful for visualising rotations
    - Commonly for robots, sensors, vehicle, planes
- 3 Variables for 3 degrees of freedom



- Different conventions though
- 3 Rotation Axes for 3 rotations
    - R_1, R_2, R_3-> X, Y, Z
- e.g. for rotation around x-axis:

- Rotation: $(\alpha, 0, 0)$



$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix}$$

$$\begin{aligned} x' &= x \\ y' &= y\cos\alpha - z\sin\alpha \\ z' &= y\sin\alpha + z\cos\alpha \end{aligned}$$

- The **Order** Makes a **Difference**

$$R_A(\alpha, \beta, \gamma) = R_3(\gamma)R_2(\beta)R_1(\alpha)$$

$$\begin{bmatrix} \cos\gamma\cos\beta & -\sin\gamma\cos\alpha + \cos\gamma\sin\beta\sin\alpha & \sin\gamma\sin\alpha + \cos\gamma\sin\beta\cos\alpha \\ \sin\gamma\cos\beta & \cos\gamma\cos\alpha + \sin\gamma\sin\beta\sin\alpha & -\cos\gamma\sin\alpha + \sin\gamma\sin\beta\cos\alpha \\ -\sin\beta & \cos\beta\sin\alpha & \cos\beta\cos\alpha \end{bmatrix}$$

$$R_B(\alpha, \beta, \gamma) = R_1(\alpha)R_2(\beta)R_3(\gamma)$$

$$\begin{bmatrix} \cos\gamma\cos\beta & -\sin\gamma\cos\beta & \sin\beta \\ \cos\gamma\sin\beta\sin\alpha + \sin\gamma\cos\alpha & -\sin\gamma\sin\beta\sin\alpha + \cos\gamma\cos\alpha & -\cos\beta\sin\alpha \\ -\cos\gamma\sin\beta\cos\alpha + \sin\gamma\sin\alpha & \sin\gamma\sin\beta\cos\alpha + \cos\gamma\sin\alpha & \cos\beta\cos\alpha \end{bmatrix}$$

Rotation Matrix to Euler Angles

- Given R: Compute (α, β, γ)
- Important: Specification for the <u>convention</u> of (α, β, γ)
  - Often 3-2-1: Yaw-Pitch-Roll OR
  - 1-2-3: Roll-Pitch-Yaw (ω, ϕ, κ)
- **Assumptions**:
  - Known axes and order of rotations
  - No **singularity** in Euler angle representation

$$R_A(\alpha,\beta,\gamma) = R_3(\gamma)R_2(\beta)R_1(\alpha) =$$
$$\begin{bmatrix} \cos\gamma\cos\beta & -\sin\gamma\cos\alpha + \cos\gamma\sin\beta\sin\alpha & \sin\gamma\sin\alpha + \cos\gamma\sin\beta\cos\alpha \\ \sin\gamma\cos\beta & \cos\gamma\cos\alpha + \sin\gamma\sin\beta\sin\alpha & -\cos\gamma\sin\alpha + \sin\gamma\sin\beta\cos\alpha \\ -\sin\beta & \cos\beta\sin\alpha & \cos\beta\cos\alpha \end{bmatrix}$$

If $\cos\beta \neq 0$:

- $\alpha = atan2(R_{32}, R_{33})$
- $\beta = atan2(-R_{31}, \sqrt{R_{32}^2 + R_{33}^2})$
- $\gamma = atan2(R_{21}, R_{11})$

If $\cos\beta = 0$:

$$R_A = \begin{bmatrix} 0 & -\sin(\gamma-\alpha) & \cos(\gamma-\alpha) \\ 0 & \cos(\gamma-\alpha) & \sin(\gamma-\alpha) \\ -1 & 0 & 0 \end{bmatrix}$$

Singularity (Gimbal Lock)

- $\beta = \pm 90°$, but α, γ not determinable
- Singularity at $\beta = 90°$
- Illustration: Makes 3rd rotation axis equal to first one (except sign)
  - Thus: Only difference $\gamma - \alpha$ inferrable, but not $\alpha,\gamma

$\rightarrow$ **Discontinuity!**

- Rotation angles limited to $[0, 2\pi)$
- "Wrap around"
- Values at borders can lead to parameter jumps
  - Problematic in state estimation

Axis-Angle Representation

- Result from Euler's Theorem
  - Every composition of 3D rotations expressable as
  - Single Rotation around Single rotation axis
- From Euler angles to Single rotation
- 4 parameters (3+1) plus 1 constraint
  - Use Vector to encode **Rotation Axis** and
  - One Scalar for Rotation Angle
- $\theta, \boldsymbol{r} = [r_1, r_2, r_3]^T$ with $||\boldsymbol{r}|| = 1$
- Alternative Minimal Representation (3 parameters):

- $\boldsymbol{r} = [r_1, r_2, r_3]^T$ with $||\boldsymbol{r}|| = \theta$

Axis-Angle to Rotation Matrix

- Rodrigues Formula:

$$R_{r,\theta} = \boldsymbol{I}_3 + \sin\theta \boldsymbol{S}_r + (1 - \cos\theta)\boldsymbol{S}_r^2$$

- With $\boldsymbol{S}_r \doteq \begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix}$
  - Normalised:
  - $R_{\boldsymbol{\theta}} = \boldsymbol{I}_3 + \frac{\sin|\boldsymbol{\theta}|}{|\boldsymbol{\theta}|}\boldsymbol{S}_\theta + \frac{1-\cos|\boldsymbol{\theta}|}{|\boldsymbol{\theta}|^2}S_\theta^2$
    - with $\boldsymbol{\theta} = \theta\boldsymbol{r}$

Rotation Matrix to Axis-Angle

- $\theta = atan2(|\boldsymbol{\alpha}|, trR - 1)$
- with:
  - $\boldsymbol{\alpha} = -\begin{pmatrix} r_{23} - r_{32} \\ r_{31} - r_{13} \\ r_{12} - r_{21} \end{pmatrix} = 2\sin\theta\boldsymbol{r}$
  - $tr(R) = r_{11} + r_{22} + r_{33} = 1 + 2\cos\theta$
- 3 cases for axis:
  1. $\theta = 0 \leftrightarrow R = I$: Singularity
  2. $\theta = \pi$: Sign of Axis is irrelevant:
     - $\boldsymbol{r}\boldsymbol{r}^T = \frac{R+I}{2}$
     - r given by any column of R
  3. Otherwise $\boldsymbol{R} = \frac{\boldsymbol{a}}{|\boldsymbol{a}|}$

Quaternions

- Complex Number with 3-D complex component
- Only 4-parameter representation for rotation that:
  - Unique, except of the sign
  - No singularities
  - No discontinuities
  - *Almost minimal*

Definition

- Quaternion $\mathbf{q} = \begin{bmatrix} q \\ \boldsymbol{q} \end{bmatrix}$ is 4 D Vector with $\boldsymbol{q} = [q_1, q_2, q_3]^T$
- Can be interpreted as
  1. Complex number
     - 1D real number q and
     - 3D complex part $\mathbf{q} = q + q_1 i + 1_2 j + q_3 k$
       - with 3 complex units i, j, k
       - for which holds $i^2 = j^2 = k^2 = ijk = -1$
  2. Tuple
     - $\mathbf{q} = (q, \boldsymbol{q})$
     - Follow rules of certain algebra

Addition

- Element-wise sum (intuitively)

$$\begin{aligned} \mathbf{p} &= \mathbf{q} + \mathbf{r} \\ (p, \boldsymbol{p}) &= (q + r, \boldsymbol{q} + \boldsymbol{r}) \\ \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix} &= \begin{bmatrix} q_0 + r_0 \\ q_1 + r_1 \\ q_2 + r_2 \\ q_3 + r_3 \end{bmatrix} \end{aligned}$$

Multiplication

- Sum/difference between cross and dot-product
- Due to cross-product: Not commutative
- Unit quaternion is quaternion $\mathbf{q}$ with $|\mathbf{q}| = 1$

$$\begin{aligned} \mathbf{p} &= \mathbf{q}\mathbf{r} \\ (p, \boldsymbol{p}) &= (qr - \boldsymbol{q} \cdot \boldsymbol{r}, r\boldsymbol{q} + q\boldsymbol{r} + \boldsymbol{q} \times \boldsymbol{r}) \\ \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix} &= \begin{bmatrix} q_0 r_0 - q_1 r_1 - q_2 r_2 - q_3 r_3 \\ q_1 r_0 + q_0 r_1 + q_2 r_3 - q_3 r_2 \\ q_2 r_0 + q_0 r_2 - q_1 r_3 + q_3 r_1 \\ q_3 r_0 + q_0 r_3 + q_1 r_2 - q_2 r_1 \end{bmatrix} \end{aligned}$$

Inverse

- Inverse $\mathbf{q}^{-1}$ of quaternion
  - with conjugate $\mathbf{q}^*$:

$$\mathbf{q}^{-1} = \frac{\mathbf{q}^*}{|\mathbf{q}|^2}$$
$$= \frac{\begin{bmatrix} q \\ -\boldsymbol{q} \end{bmatrix}}{q_0^2 + q_1^2 + q_2^2 + q_3^2}$$

Rotations with Quaternions

$$\mathbf{q} = \begin{bmatrix} q \\ \boldsymbol{q} \end{bmatrix} = \begin{bmatrix} \cos(\theta/2) \\ \sin(\theta/2)\boldsymbol{r} \end{bmatrix}$$

- With normalised rotation axis:
  - $\boldsymbol{r} = [r_1, r_2, r_3]^T$ with $||\boldsymbol{r}|| = 1$
- Is Similar to Axis-Angle
  - $\theta, \boldsymbol{r} = [r_1, r_2, r_3]^T$ with $||\boldsymbol{r}|| = 1$

Example: Quaternion Rotation

- A rotation of θ around $\boldsymbol{r}$ with

$$\theta = 30^\circ \qquad \boldsymbol{r} = \begin{bmatrix} 0.6 \\ 0.8 \\ 0.0 \end{bmatrix}$$

- Equation: $\mathbf{q} = \begin{bmatrix} q \\ \boldsymbol{q} \end{bmatrix} = \begin{bmatrix} \cos(\theta/2) \\ \sin(\theta/2)\boldsymbol{r} \end{bmatrix}$

- Result (in different notations):

$$\mathbf{q} = \begin{bmatrix} \cos(15^\circ) \\ \sin(15^\circ) \begin{bmatrix} 0.6 \\ 0.8 \\ 0.0 \end{bmatrix} \end{bmatrix}$$

$$\mathbf{q} = (\cos(15^\circ), \sin(15^\circ)(0.6, 0.8, 0.0))$$
$$\mathbf{q} = \cos(15^\circ) + \sin(15^\circ)(0.6\,i + 0.8\,j + 0.0\,k)$$

Executing a Rotation

- Perform rotation of vector part $\boldsymbol{p}$ of $p = (0, \boldsymbol{q})$
- Through left and right multiplication by $\boldsymbol{q} = (\cos(\theta/2), sin(\theta/2)\boldsymbol{r})$

$$\mathbf{p}' = \mathbf{q}\mathbf{p}\mathbf{q}^{-1}$$

Composition

- (Multiple) Rotations as composition of quaternion multiplications

$$\mathbf{p}' = \mathbf{q}'\mathbf{p}\mathbf{q}'^{-1} \qquad \mathbf{p}'' = \mathbf{q}''\mathbf{p}'\mathbf{q}''^{-1}$$

$$\begin{aligned}
\mathbf{p}'' &= \mathbf{q}''\,(\mathbf{q}'\,\mathbf{p}\,\mathbf{q}'^{-1})\,\mathbf{q}''^{-1} \\
&= (\mathbf{q}''\mathbf{q}')\,\mathbf{p}\,(\mathbf{q}'^{-1}\mathbf{q}''^{-1}) \\
&= \underbrace{(\mathbf{q}''\mathbf{q}')}_{\mathbf{q}}\,\mathbf{p}\,\underbrace{(\mathbf{q}''\mathbf{q}')^{-1}}_{\mathbf{q}^{-1}} \\
&= \mathbf{q}\mathbf{p}\mathbf{q}^{-1}
\end{aligned}$$

- composite rotation $\mathbf{q} = \mathbf{q}''\mathbf{q}'$
  - Equal to $R_{\mathbf{q}} = R_{\mathbf{q}''}R_{\mathbf{q}'}$
- of rotation quaternions $\mathbf{q}'', \mathbf{q}'$
- Execute rotation by

$$\mathbf{p}''\mathbf{q}\mathbf{p}\mathbf{q}^{-1}$$

Quaternion and Rotation Matrix

- A unit quaternion can be transformed into a rotation matrix by

$$R_{\mathbf{q}} = \begin{bmatrix} 1-2\,(q_2^2+q_3^2) & 2\,(q_1q_2-q_0q_3) & 2\,(q_1q_3+q_0q_2) \\ 2\,(q_2q_1+q_0q_3) & 1-2\,(q_1^2+q_3^2) & 2\,(q_2q_3-q_0q_1) \\ 2\,(q_3q_1-q_0q_2) & 2\,(q_3q_2+q_0q_1) & 1-2\,(q_1^2+q_2^2) \end{bmatrix}$$

- Conversion in the other direction

$$\begin{aligned}
q_0 &= \frac{1}{2}\sqrt{1+\mathrm{tr}R} \\
q_1 &= (R_{32}-R_{23})/(4q_0) \\
q_2 &= (R_{13}-R_{31})/(4q_0) \\
q_3 &= (R_{21}-R_{12})/(4q_0)
\end{aligned}$$

Summary

3D Rotation Parameterisations

Representation	# Parameters	Unique	Readable	Singularities
Rotation Matrix	9	✅	❌	❌
Euler Angles	3	❌	✅	✅
Rotation Axis and Angle	4(3)	(❌)	(✅)	(✅)
Quaternion	4(3)	✅	❌	❌

- Different Representations differ w.r.t.:
  - Readability
  - Singularities
  - Discontinuities
  - Minimal vs. overparameterised
  - Uniqueness
  - Simplicity of rotation composition
- -> No Best representation
- **Rotation Matrices**: **Exchange Format**
- **Euler Angles**: **Visualisation**
- **Quaternions**: **State Estimation**
- **Angle-Axis**: **Readability and Link to Quaternions**